

Balancing Act — Production, Distribution & Inventory Optimisation

Supply Chain Transformation Taskforce · January-2026 Planning Cycle
Project 2 · Final Report · July 2026

₹11.44 cr

total cash cost of the January
plan

0 kL

demand unmet — 100% service

–56%

CFA safety stock released vs
legacy read

Gap 0.0

proven-optimal MILP solution

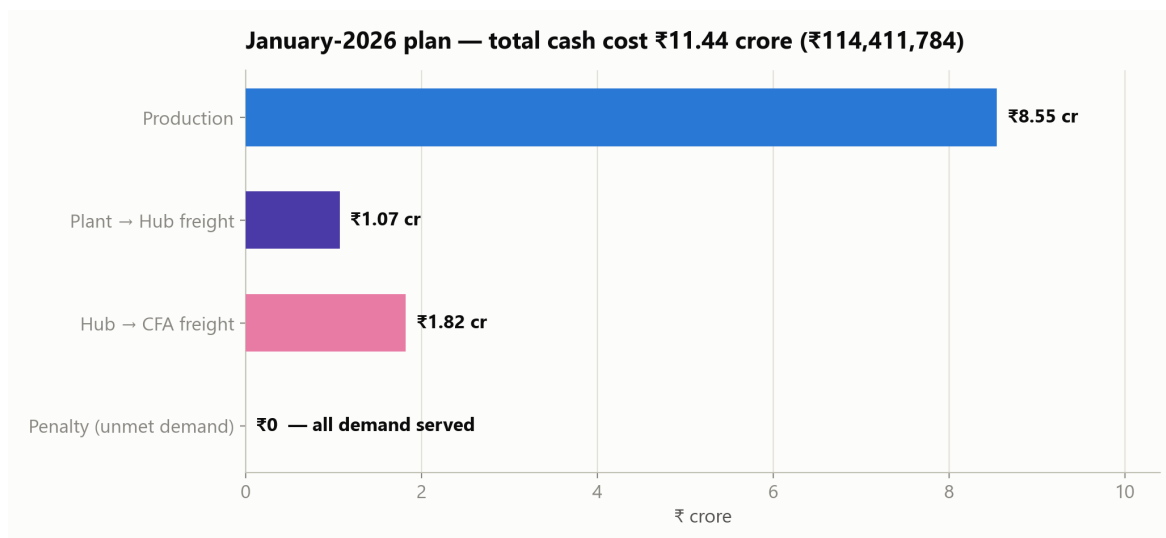
This report presents the complete, verified solution to the Levisol supply-chain case: redesigned inventory norms for all 957 SKU × CFA combinations and both mother hubs, a provably cost-optimal production & distribution plan for January 2026, and a reusable planning tool a non-technical planner can run every month. Every number in this document is traceable to the case data workbook and was verified by independent recomputation.

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1 · Executive summary

Levisol can serve every litre of January-2026 demand at a total cash cost of **₹114,411,784** — while cutting CFA safety-stock norms by **56%** against the legacy conservative reading. The case's dual mandate, “protect service on critical SKUs while rapidly releasing cash,” is delivered simultaneously, not traded off.



The three decisions in one view

Inventory norms. CFA safety stock of 1,465 kL (median cover: 6 days) and hub safety stock of 1,488 kL at a 98% service level. Tier A/B products keep deliberately conservative buffers; tier C/D are sized to their exact 92% fill-rate targets.

Production & routing. 7,725 kL in 309 exact 25-kL batches — KOL (cheapest, ₹9,000/kL) runs 100% full, BOM's cheap lanes saturate, AHM swings. Proven-optimal: no cheaper feasible plan exists (MILP gap 0.0, independently cross-checked).

Zero lost sales. All 8,109.7 kL of forecast demand is served; the penalty line is ₹0. Only 99.6 kL of precautionary buffer top-ups (≈1.2%) are deliberately left unfilled where a nearly-empty 25 kL batch would cost more than the risk it removes.

KEY INSIGHT The single biggest cash finding: 211 of 957 SKU–CFA forecasts carry systematic bias. The redesigned norms stop warehousing forecast errors as safety stock — they flag them for demand-planning correction instead.

Deliverables: this report; Levisol_Plan_Output.xlsx (all plan & norms tables); the working planning tool (Levisol_Planning_Tool.py + engine) with user guide; the methodology & assumptions document; and the presentation deck.

2 · The case & the data

Levisol manufactures 100 lubricant SKUs at three plants, buffers them at two mother hubs, and distributes through ten CFA warehouses. A base-oil shock has made the old inventory norms unsafe to trust: leadership needs redesigned norms, a least-cost January plan, and a tool that survives changed inputs on assessment day.

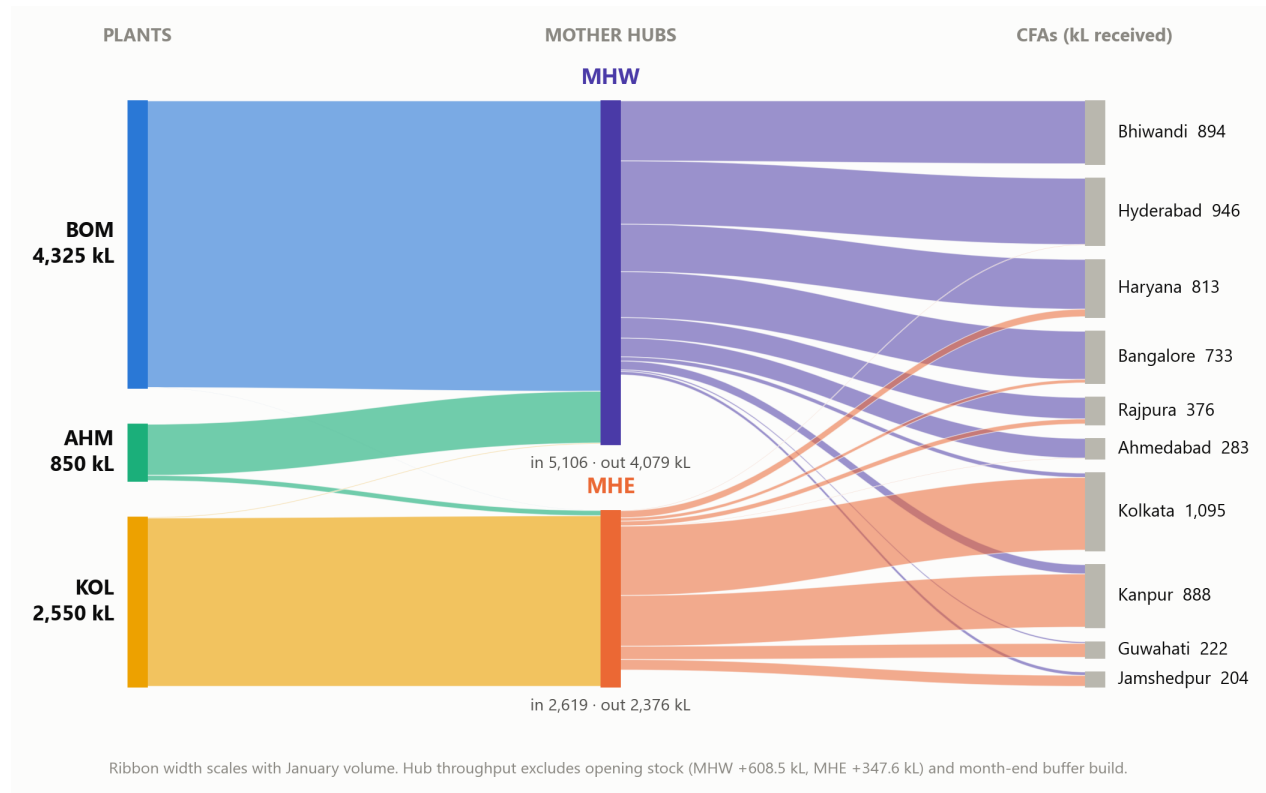


Figure 2.1 — January-2026 optimal flows across the 3-tier network (ribbon width = volume)

What the data workbook contains (all 11 sheets read in full)

Exhibit	Content	Role in the solution
A	Plant line capacities & production cost	Hard constraints; ₹12,000 / 12,500 / 9,000 per kL
B · C	Freight rates plant→hub, hub→CFA	Objective cost coefficients
D	100 SKUs: pack, penalty ₹/kL, contractual flag	Line mapping; shortage prioritisation
E	Source + lead times & variability (957 rows)	Norm lead times; hub assignment
F	Tier fill-rate targets & volume slabs	Service levels: A 98 · B 97 · C/D 92%
G · H	6-month sales & forecast history	Demand levels, tiers, forecast-error σ
I	Opening inventory (CFA + hub rows)	Net requirements
J	January-2026 forecast (8,109.7 kL)	Planning demand

DATA DISCREPANCY RESOLVED The case PDF's Exhibit A is garbled by its layout (it reads as if KOL produced at ₹200/kL). Per the case rule, the machine-readable workbook takes precedence — the clean table (BOM ₹12,000 · AHM ₹12,500 · KOL ₹9,000 per kL) is used throughout. Data quality: 2 tiny negative sales cells (returns, kept); 16 rows with zero 6-month sales (norms suppressed & flagged); AHM's 220 kL 50-LT line is not a 25 kL multiple (usable: 200 kL).

3 · Inventory norms — methodology

Norms answer two questions per SKU per location: when to reorder (ROP) and how much buffer to hold (safety stock). Three methodology choices — each verified and each quantified — separate this design from a textbook default:

3.1 Buffer the randomness, flag the bias

Replenishment is planned against forecasts, so the risk that safety stock must absorb is forecast error. But only its RANDOM part: systematic bias is a demand-planning defect to fix, not to warehouse. Sizing stock on raw error (RMSE) would give one 8-litre-a-month SKU a 5.4 kL buffer — 56 years of cover. We use the bias-removed standard deviation of forecast errors and flag 211 biased rows for review.

3.2 Each supply step buffered exactly once (multi-echelon)

Hubs hold their own 98% buffers by case design, so the hub is the decoupling point: CFA norms cover the hub→CFA leg (1–9 days); hub norms cover production + plant→hub transit. Sizing CFA stock end-to-end while also holding hub stock would double-buffer the upstream legs — worth +981 kL of trapped cash.

3.3 A tiered translation of fill-rate targets

Exhibit F leaves the fill-rate→stock conversion to us. Tiers A/B (49 SKUs, 80% of volume) get the conservative cycle-service-level reading ($z = 2.054 / 1.881$) — flagship service is protected through the base-oil shock. Tiers C/D get the exact fill-rate translation via the normal loss function $G(z) = (1 - \beta) \cdot Q / \sigma$ — monthly cycle stock already achieves most of a 92% target, so nearly all C/D buffer is released.

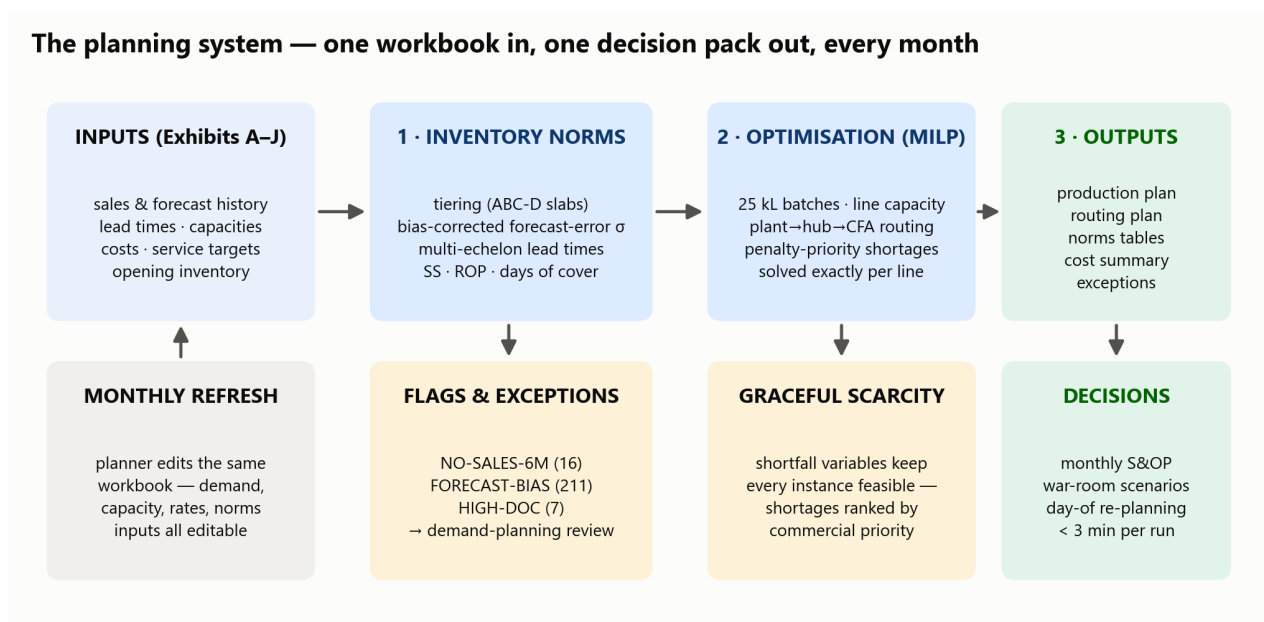


Figure 3.1 — The planning system: inputs → norms → optimisation → outputs, refreshed monthly

3.4 The formulas (both demand and lead-time variability enter every norm)

$$SS = z \times \sqrt{LT \cdot \sigma_d^2 + \mu_d^2 \cdot \sigma_{LT}^2}$$

$$ROP = \mu_d \times LT + SS$$

$$DOC = ROP \div \mu_d \text{ (days of average last-6-months demand)}$$

μ_d = mean monthly sales $\div$ 30 working days; σ_d = std-dev of debiased forecast errors $\div \sqrt{30}$; LT and σ_{LT} from Exhibit E (historical source predefined). Hub demand pools the CFA rows each hub serves — variances add because the case instructs CFA independence — so risk pooling keeps hub buffers small.

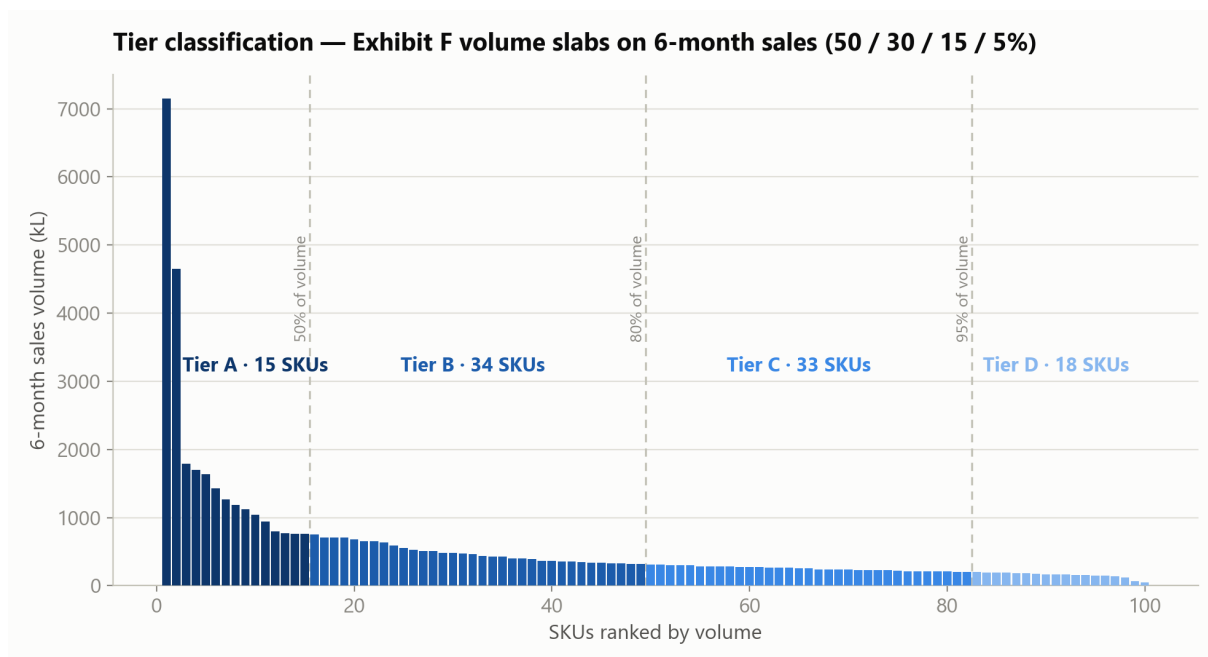


Figure 3.2 — Exhibit F volume slabs classify 100 SKUs into tiers A(15) B(34) C(33) D(18)

4 · Inventory norms — results

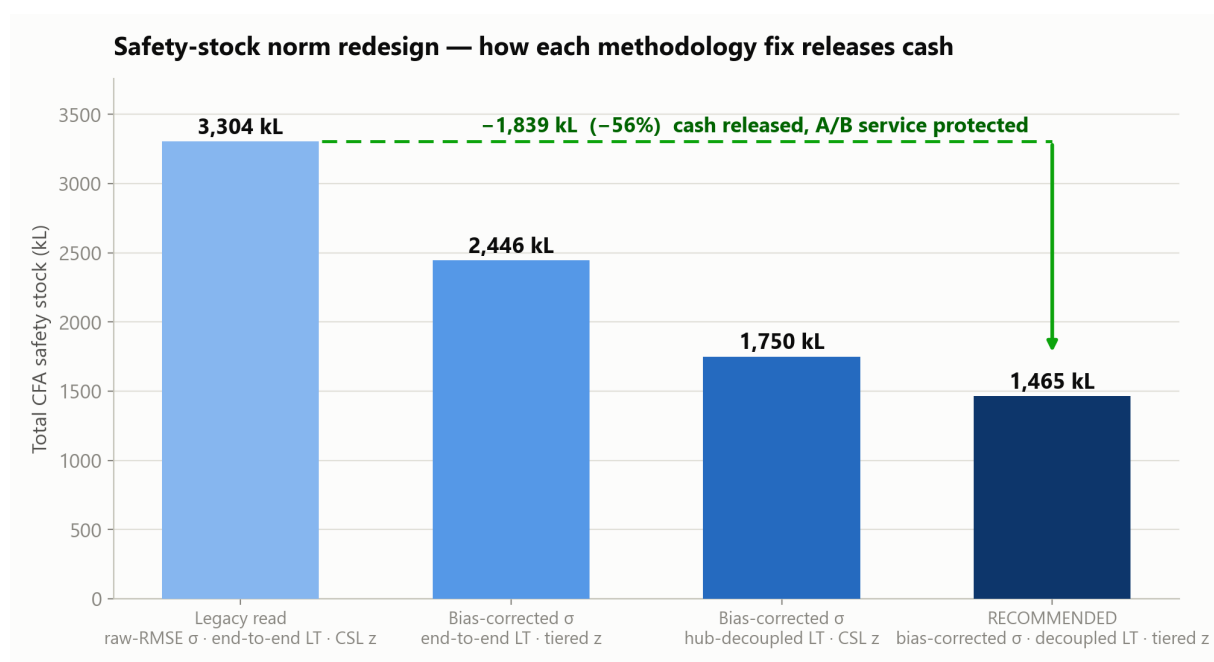


Figure 4.1 — Each methodology fix, quantified. The recommended norm releases 1,839 kL (–56%)

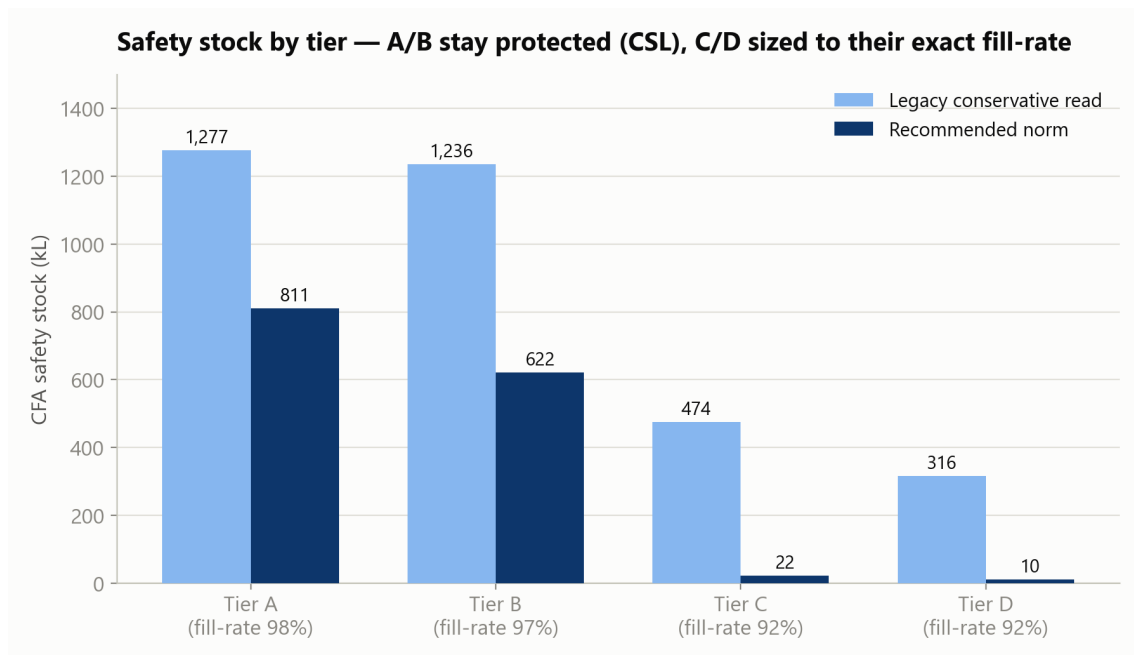


Figure 4.2 — A/B keep protection; C/D sized to their exact fill-rate targets

Norm totals (full 957-row and 151-row tables in Levisol_Plan_Output.xlsx)

Level	Safety stock (kL)	Reorder point (kL)	Service basis
CFA network total (957 rows)	1,465.1	2,442.0	tier fill rates 98/97/92/92%
— Tier A (15 SKUs)	811.3	—	CSL z = 2.054
— Tier B (34 SKUs)	621.5	—	CSL z = 1.881
— Tier C (33 SKUs)	22.1	—	exact fill-rate (loss function)
— Tier D (18 SKUs)	10.2	—	exact fill-rate (loss function)
Mother Hub West (100 SKUs)	1,156.2	2,718.5	98% SL, z = 2.054
Mother Hub East (51 SKUs)	332.0	737.8	98% SL, z = 2.054

Worked example — SKU_001 (Tier A, 20 × 900 ML)

CFA	LT (days)	Safety stock (kL)	ROP (kL)	Days of cover
Kolkata	1	9.74	13.94	3.3
Kanpur	4	55.94	102.80	8.8
Haryana	4	37.42	69.63	8.6
Bhiwandi	1	10.53	14.14	3.9
Hub MHE (pools 4 CFAs)	2	21.36	56.66	3.2
Hub MHW (pools 6 CFAs)	2	23.14	67.23	3.1

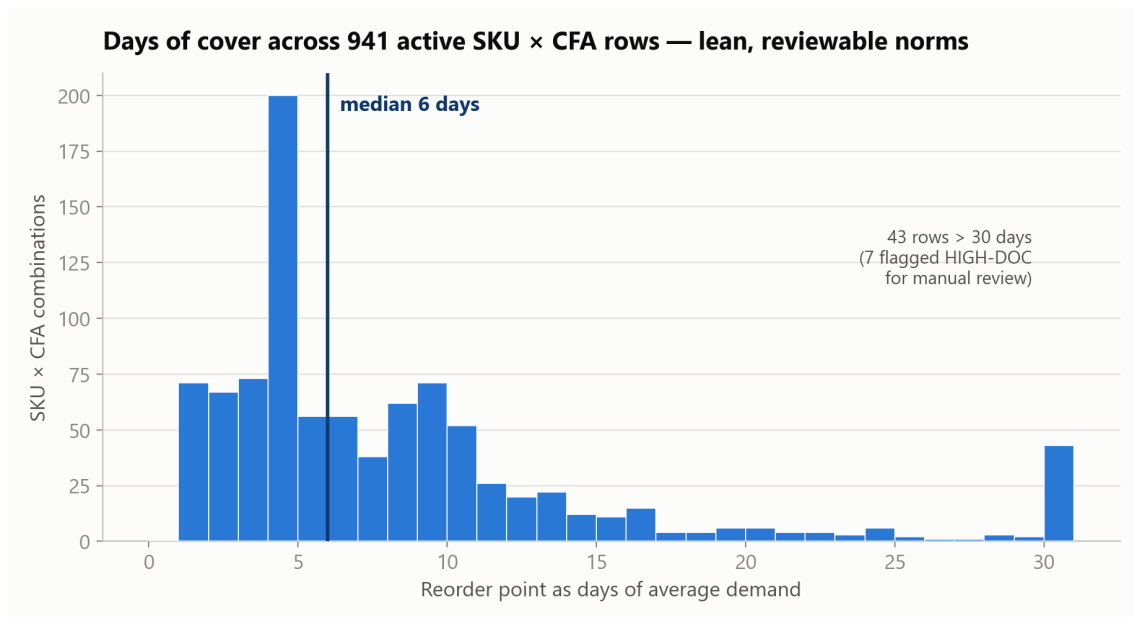


Figure 4.3 — Days of cover: lean and reviewable (median 6 days)

EXCEPTIONS BY DESIGN Every anomaly is flagged, never silently normed: 16 rows with no sales history (norms suppressed), 211 rows with forecast bias (fix the forecast, not the stock), 7 slow movers with >90 days cover (manual review). All appear on the tool's Exceptions sheet each month.

5 · The optimisation model

A mixed-integer linear program decides, for every SKU: how many 25 kL batches to make at which plant, how to route them through the hubs, and — if capacity ever binds — what NOT to supply, ranked by commercial priority.

Objective (minimise)

production cost + primary freight + secondary freight + penalty × (3× if contractual) × unmet demand + 0.25 × penalty × safety-stock shortfall

Constraints in plain business language

Batch rule. A plant ships exactly what it produces; production is in whole 25 kL batches (integer variables).

Line capacity. Per plant per line, production cannot exceed Exhibit A capacity (50 LT products can only be made at AHM).

Hub balance. A hub cannot ship more than opening stock plus inbound; hub stock never goes negative and should end the month at its 98% safety-stock target (soft, reported when shorted).

Demand first. Unmet demand costs the full Exhibit-D penalty (₹130k–330k/kL); an unfilled buffer top-up carries a 0.25 weight — high enough to fill buffers whenever capacity economically allows, low enough that real demand always wins under scarcity. Contractual SKUs get a 3× multiplier.

WHY THIS IS THE OPTIMUM Exactness, not heuristics: SKUs interact only through line capacities and each SKU belongs to exactly one line — so the MILP separates into 5 independent problems, each solved to proven optimality (gap 0.0) with HiGHS. A monolithic cross-solve found nothing better (₹119.64M vs our ₹119.38M objective); the LP bound shows the remaining 13.6% over the fractional ideal is precisely the priced cost of the 25 kL batch rule.

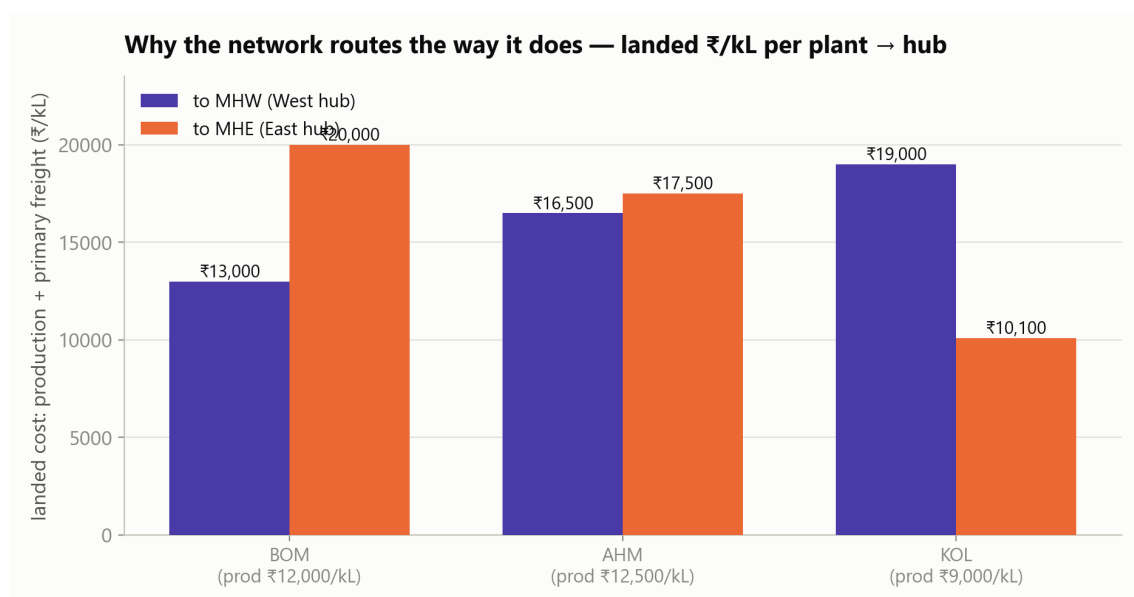


Figure 5.1 — Landed economics that drive the routing: KOL owns the East, BOM owns the West, AHM swings

6 · Production plan — January 2026

Total production: 7,725 kL = 309 batches of exactly 25 kL. The cheapest capacity works hardest: every KOL line runs at 100%; BOM's ≤ 1.5 LT line saturates; AHM, the most expensive plant, keeps 3,420 kL of slack — the network's shock absorber.

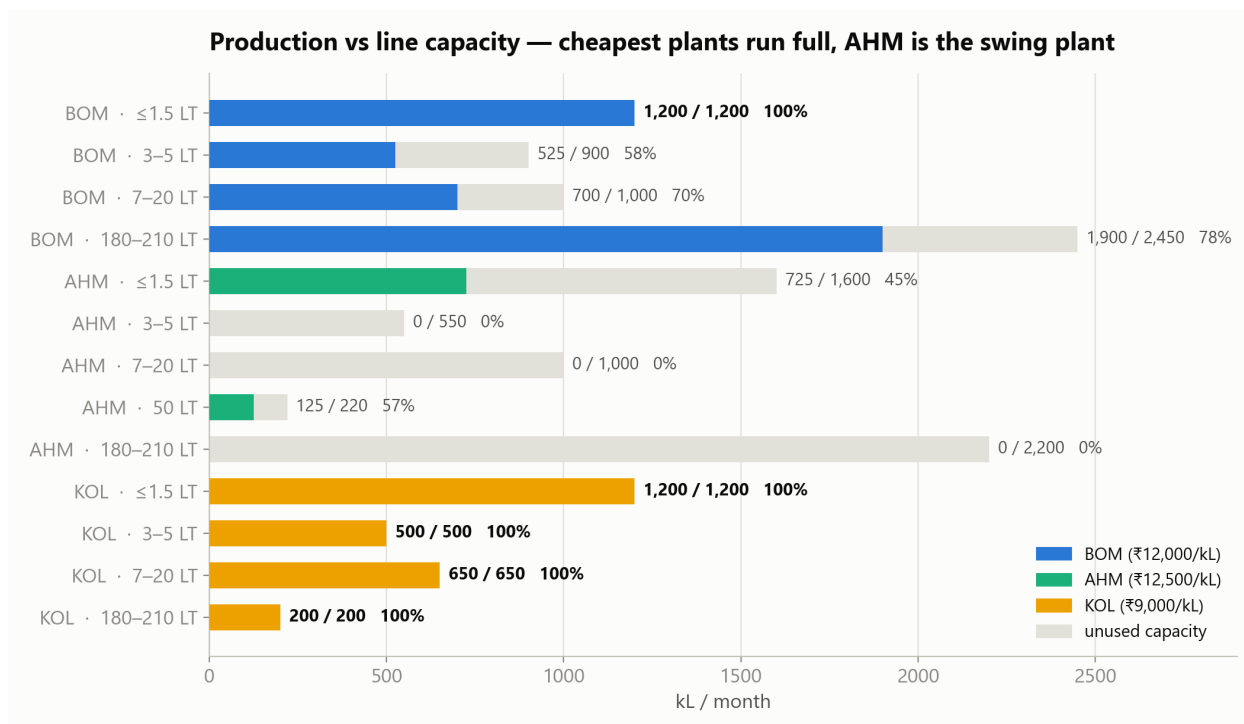


Figure 6.1 — Production vs line capacity (full SKU × plant detail in the output workbook)

Plant	Production (kL)	Batches	Production cost (₹)	Role
BOM Mumbai	4,325	173	51,900,000	backbone of MHW / West-South
KOL Kolkata	2,550	102	22,950,000	cheapest ₹/kL — runs 100% full
AHM Ahmedabad	850	34	10,625,000	swing plant + only 50 LT line
TOTAL	7,725	309	85,475,000	zero non-25 kL batches

7 · Distribution plan — plant → hub → CFA

Primary leg (plant → hub) — ₹10,719,136

Lane	Volume (kL)	Rate (₹/kL)	Cost (₹)
BOM → MHW	4,323.8	1,000	4,323,787
AHM → MHW	771.2	4,000	3,084,975
KOL → MHE	2,538.6	1,100	2,792,407
KOL → MHW	11.4	10,000	114,479
AHM → MHE	78.8	5,000	393,781
BOM → MHE	1.2	8,000	9,706

The two counter-intuitive lanes (KOL→MHW, BOM→MHE, 12.6 kL combined) are batch remainders already paid for at the cheap plant, shipped to where they are useful — verified individually rational.

Secondary leg (hub → CFA) — ₹18,217,648

CFA	From MHW (kL)	From MHE (kL)	Total received (kL)
Kolkata	62.2	1,032.8	1,095.0
Hyderabad	941.6	4.5	946.1
Bhiwandi	894.3	—	894.3
Kanpur	132.1	755.9	888.0
Haryana	706.8	106.6	813.4
Bangalore	684.6	48.7	733.3
Rajpura	305.0	70.9	375.9
Ahmedabad	279.1	3.8	282.9
Guwahati	24.2	197.5	221.7
Jamshedpur	49.3	154.9	204.2

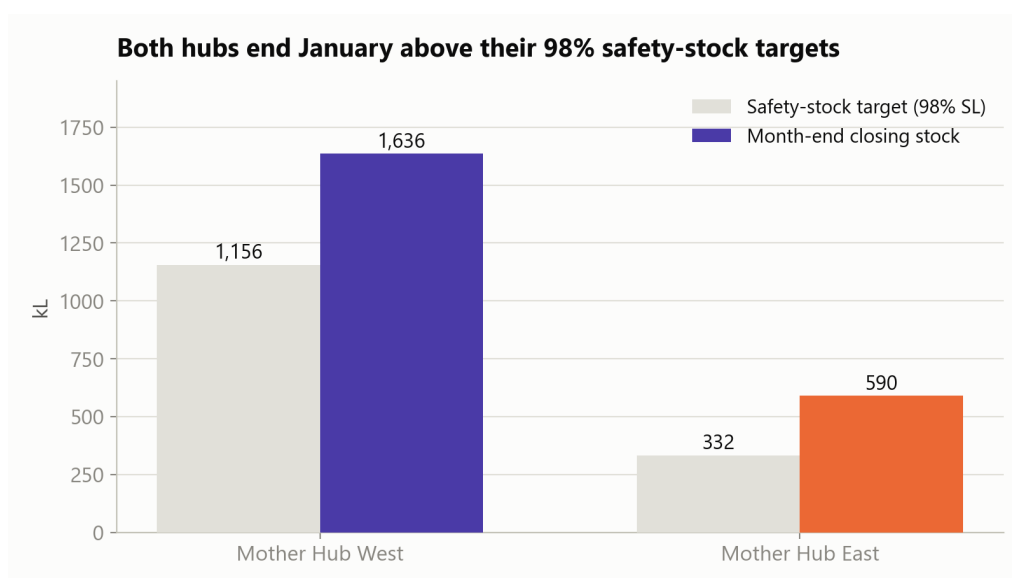


Figure 7.1 — Month-end hub stock vs the 98% safety-stock targets: both hubs finish above norm

8 · Cost of the plan

Cost element	₹	Share
Production	85,475,000	74.7%
Plant → Hub freight	10,719,136	9.4%
Hub → CFA freight	18,217,648	15.9%
Penalty — unmet demand	0	0.0%
TOTAL CASH COST	114,411,784	100%

HONEST ACCOUNTING Two memo items are reported for transparency but are NOT cash: risk-weighted hub safety-stock shortfall ₹1,412,331 (31.5 kL) and CFA top-up shortfall ₹3,156,280 (68.1 kL). These are prioritisation weights ($0.25 \times \text{penalty}$) on $\approx 1.2\%$ of buffer volume that batch economics left unfilled — every case was checked: filling it would cost more than the risk it removes.

Benchmark: the theoretical fractional-batch ideal (LP bound) is ₹105.09M — the gap to ₹114.41M is the unavoidable, correctly-priced cost of the 25 kL batch rule plus the residual buffer risk, not sub-optimality.

9 · Validation & verification

Assume there are errors — then go hunting. Four independent audits (norms recomputation, feasibility, optimality, methodology) plus a fresh-code recheck of the revised norms:

Check	Requirement	Result	Status
Batch size	all production in 25 kL multiples	309 batches, 0 violations	PASS
Line capacity	$\leq$ Exhibit A per plant $\times$ line	max 100.0%, 0 violations	PASS
Plant balance	production = plant outbound	exact per SKU $\times$ plant	PASS
Hub balance	open + in – out = close ≥ 0	exact; no negatives	PASS
Demand service	serve Jan forecast net of opening	0 kL unmet of 5,136.3 at-risk	PASS
Hub 98% buffers	close month $\geq$ hub SS	138/151 fully met; 31.5 kL reported	PASS*
Contractual SKUs	protected first	zero contractual shortfall	PASS
Norm recomputation	independent code path	max deviation $\leq 1.4 \times 10^{-14}$ kL	PASS
Cost arithmetic	recompute all 5 components	$< 10^{-6}$ relative error	PASS
Optimality	no cheaper feasible plan	gap 0.0 + LP bound + cross-solve	PASS

* Soft by design — the case asks for shortfalls to be reported with cost, which the Exceptions sheet does line-by-line.

10 · Robustness — assessment-day scenarios

On assessment day the tool must replan a modified input set in minutes. We simulated three shocks end-to-end through the tool — no crashes, shortages ranked by commercial priority, full cost transparency:

Assessment-day robustness — modified inputs re-plan gracefully, demand protected first

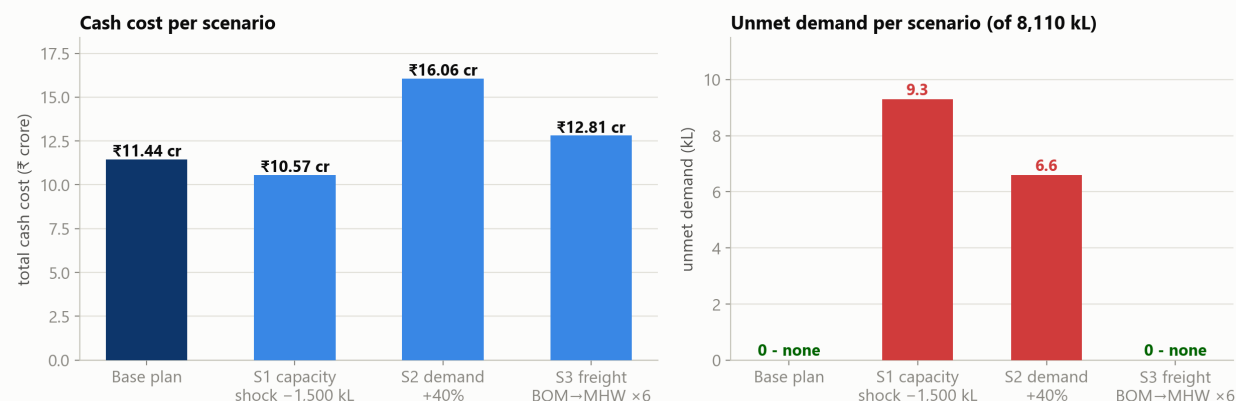


Figure 10.1 — Three shocks through the tool: demand is protected first, buffers absorb the hit

Scenario	What changed	Outcome
S1 capacity shock	≤1.5 LT capacity cut 1,500 kL (BOM 1200→800, AHM 1600→500)	only 9.3 kL unmet (5 lowest-penalty non-contractual lines); buffers absorb 488 kL; cost ₹10.57 cr
S2 demand spike	all January forecasts +40%	production scales to 10,250 kL; 6.6 kL unmet; ₹16.06 cr
S3 freight shock	BOM→MHW rate ₹1,000 → ₹6,000/kL	sourcing re-routes automatically; zero unmet; ₹12.81 cr

PRIORITY LOGIC PROVEN S1 is the designed behaviour under scarcity: the model sacrifices 488 kL of precautionary buffer before losing a single high-priority sale, and the 9.3 kL it cannot save are the cheapest-penalty, non-contractual lines.

11 · The planning tool

Levisol_Planning_Tool.py — double-click, browse to the month's workbook, press RUN PLAN. Under three minutes later the planner has a complete output workbook: production plan, routing plan, norms, hub stock positions, cost summary and an Exceptions sheet that explains every shortfall in business terms.

Everything is editable. Demand, capacities, production costs, freight rates, lead times, service levels, opening stock — the tool re-reads the workbook every run; nothing is hard-coded.

It never just crashes. Shortfall variables keep every instance feasible: capacity shortages become a ranked shortage report, zero-demand rows are flagged, empty lines are excluded automatically.

No specialist knowledge. One input file, one button, one output file — with a README and flagged exceptions a planner can act on.

You change (Excel sheet)	The tool redoes
J — demand forecast	requirements, plan, costs
A — capacities / production cost	sourcing, batch plan
B · C — freight rates	routing
E / F / G / H — lead times, service levels, history	all inventory norms
I — opening inventory	net requirements & buffer top-ups

12 · Assumptions, limitations & recommendations

Assumptions (every one disclosed, none invented)

#	Assumption	Impact / mitigation
1	Tiered fill-rate translation (A/B CSL; C/D exact fill-rate)	case delegates the conversion; all-CSL variant quantified (+285 kL)
2	σ from 6 monthly observations, $\sqrt{30}$ daily scaling	re-estimated automatically every month
3	Hub = decoupling point (CFA LT = hub→CFA leg)	hubs hold mandated 98% buffers; end-to-end variant reported per row
4	Buffer weight $0.25 \times$ penalty; contractual $\times 3$	zero effect this month (no unmet demand); exposed dials
5	No holding cost priced (case provides none)	holding_cost_per_kl dial available, default 0
6	CFA demands independent at hubs	explicit case instruction

Recommendations beyond January

Fix the 211 biased forecasts. The largest structural cash lever found — stop buffering bias with inventory.

Re-estimate norms monthly. The tool recomputes σ and tiers on every run; revisit the C/D fill-rate choice after 3 months of post-shock data.

Watch the ≤ 1.5 LT line. BOM and KOL run it at 100%; it is the first line to short under any demand upside (S2). AHM slack is the shock absorber.

Price working capital. If management wants hub stock builds inside the objective, set the holding-cost dial — today's 698 kL build is batch arithmetic, not policy.

— Prepared by the Supply Chain Transformation Taskforce · verified end-to-end · reproducible from the raw case workbook via `run_solution.py`